William Curran

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SolidWorks Certified Associate | SEM (Scanning Electron Microscope) Trained & Certified

EDUCATION

University of Massachusetts Lowell — B.S., Mechanical Engineering *Expected May 2028*

GPA: 3.4

- Relevant Coursework: Thermodynamics, Dynamics, Statics & Mechanics of Materials, Physics I II & III, Multivariable Calculus, Computer-Aided Design, Manufacturing Lab, Materials Science for Engineers

EXPERIENCE

MINI LAB | Manufacturing Innovation and Neural Interface Lab, Undergraduate Researcher *Winter 2026–Present*

- Led two high school interns on the NEF project and regularly present research findings to the lab
- Assist PhD, master's, and undergraduate colleagues with an automated BMI assembly machine and methods for inserting wires of 50 µm or less into a mock brain

Neural Electrode Fabricator (NEF)

- Engineered NEF, a precision laser system that sharpens neural microelectrode wire, including 7 µm carbon fiber, and strips insulation to an exact length, leaving the conductor intact
- Supplies the sharpening and stripping parameters for the lab's automated BMI assembler, whose chips are tested in animal models, toward BMIs that could help people with speech disabilities communicate
- Integrated the motorized stage, laser power control, and microcontroller into one electromechanical system with user-set speed, power, and pass count; characterized stable processing regimes
- Built a Python logger that records every run for repeatable, data-driven tuning

NEF Upgrade: Custom Fiber-Coupled Laser Source *Fall 2026*

- Built a custom fiber-delivered laser source for NEF, replacing the off-the-shelf module with higher power and flexible beam delivery
- Developed the driver, cooling, focusing optics, and beam dump, and calculated spot size and working distance

NERVE CENTER | UMass Lowell Robotics Research Center, Student Researcher *Fall 2026–Present*

Ankle Exosuit *Design Phase, Fall 2026–Present*

- Designing an ankle exosuit that assists walking to reduce the energy wearers spend, aimed at people with arthritis, Achilles tendonitis, or runner's knee
- Prioritizing dexterity, comfort, and smooth, non-jerky motion; developing the CAD model to refine geometry, fit, and load paths ahead of prototyping
- Planning custom footplates fitted to each wearer for comfort and effectiveness, and defining the electromechanical architecture (actuation, sensing, power) around smooth force delivery and safety

ENGINEERING PROJECTS

Industrial Robotic Arm — Fourth Design Iteration *Spring 2026–Present*

- Designed and built a 6-DOF arm for high torque and repeatable motion with custom 3D-printed planetary gearboxes
- Built custom driver electronics and power distribution around an ESP32-S3; firmware in development, natural-language control planned
- Next upgrade: adding joint sensors and a camera for sensor-guided control

Teleoperated Robotic Manipulator *Winter 2026*

- Built a 4-DOF arm that mirrors live human hand motion using Python and MediaPipe hand tracking
- Designed all links in SolidWorks and fixed torque and wobble under load by shortening the arm and stiffening the links

TECHNICAL SKILLS

- **Electromechanical Design:** Actuator sizing, gear reduction, torque and moment analysis, power distribution
- **Mechanical Design & Analysis:** SolidWorks (Certified Associate), GD&T, ANSYS FEA
- **Electrical & Embedded:** Custom control boards, ESP32, Arduino, TMC2209 drivers, PlatformIO, sensor integration
- **Manufacturing:** SLA and FDM 3D printing, CNC and lathe machining, laser micromachining
- **Software:** Python, MediaPipe, Flask/FastAPI, LLM-integrated pipelines